

Supporting Information for Molecular Vibration Explorer: an Online Database and Toolbox for Surface-Enhanced Frequency Conversion, Infrared and Raman Spectroscopy

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S1 Analytic formulas for orientation-averaging

There are four unique integrals for all the possible arrangements of the field vectors, these are given in the following. Subscripts a , b , and c denote field directions orthogonal to each other.

$$\langle |e_a \underline{\mu}'_m|^2 |e_a \underline{\alpha}'_m e_a|^2 \rangle =$$

$$\begin{aligned}
& \frac{1}{105} \left(2\alpha'_{m,xx} (12\mu'_{m,x}\alpha'_{m,xy}\mu'_{m,y} + 4\mu'_{m,z} (3\mu'_{m,x}\alpha'_{m,xz} + \mu'_{m,y}\alpha'_{m,yz})) \right. \\
& + \alpha'_{m,yy} (3(\mu'_{m,x})^2 + 3(\mu'_{m,y})^2 + (\mu'_{m,z})^2) \\
& + \alpha'_{m,zz} (3(\mu'_{m,x})^2 + (\mu'_{m,y})^2 + 3(\mu'_{m,z})^2) \\
& + 3(\alpha'_{m,xx})^2 (5(\mu'_{m,x})^2 + (\mu'_{m,y})^2 + (\mu'_{m,z})^2) \\
& + 8\alpha'_{m,xy} (2\mu'_{m,z} (\mu'_{m,x}\alpha'_{m,yz} + \alpha'_{m,xz}\mu'_{m,y}) \\
& + \mu'_{m,x}\mu'_{m,y} (3\alpha'_{m,yy} + \alpha'_{m,zz})) \\
& + 4(\alpha'_{m,xy})^2 (3((\mu'_{m,x})^2 + (\mu'_{m,y})^2) + (\mu'_{m,z})^2) \\
& + 12(\mu'_{m,x})^2 (\alpha'_{m,xz})^2 \\
& + 8\mu'_{m,z} (\mu'_{m,x}\alpha'_{m,xz} (\alpha'_{m,yy} + 3\alpha'_{m,zz}) + 3\mu'_{m,y}\alpha'_{m,yz} (\alpha'_{m,yy} + \alpha'_{m,zz})) \\
& + 16\mu'_{m,x}\alpha'_{m,xz}\mu'_{m,y}\alpha'_{m,yz} + 3(\mu'_{m,x})^2 (\alpha'_{m,yy})^2 \\
& + 2(\mu'_{m,x})^2 \alpha'_{m,yy}\alpha'_{m,zz} + 4(\mu'_{m,x})^2 (\alpha'_{m,yz})^2 + 3(\mu'_{m,x})^2 (\alpha'_{m,zz})^2 \\
& + 4(\alpha'_{m,xz})^2 (\mu'_{m,y})^2 + 3(\mu'_{m,z})^2 (4(\alpha'_{m,xz})^2 \\
& + (\alpha'_{m,yy})^2 + 2\alpha'_{m,yy}\alpha'_{m,zz} + 4(\alpha'_{m,yz})^2 + 5(\alpha'_{m,zz})^2) \\
& + 15(\mu'_{m,y})^2 (\alpha'_{m,yy})^2 + 6(\mu'_{m,y})^2 \alpha'_{m,yy}\alpha'_{m,zz} \\
& + 12(\mu'_{m,y})^2 (\alpha'_{m,yz})^2 + 3(\mu'_{m,y})^2 (\alpha'_{m,zz})^2 \Big)
\end{aligned} \tag{S1}$$

$$\langle |e_b \underline{\mu}'_m|^2 |e_a \underline{\alpha}'_m e_a|^2 \rangle =$$

$$\begin{aligned} & \frac{1}{105} \left(3 (\mu'_{m,x})^2 (\alpha'_{m,xx})^2 \right. \\ & - 4 \mu'_{m,z} (\mu'_{m,x} \alpha'_{m,xz} (3 \alpha'_{m,xx} + \alpha'_{m,yy} + 3 \alpha'_{m,zz}) + 2 \mu'_{m,x} \alpha'_{m,xy} \alpha'_{m,yz} \\ & + \mu'_{m,y} \alpha'_{m,yz} (\alpha'_{m,xx} + 3 (\alpha'_{m,yy} + \alpha'_{m,zz})) + 2 \alpha'_{m,xy} \alpha'_{m,xz} \mu'_{m,y}) \\ & - 12 \mu'_{m,x} \alpha'_{m,xx} \alpha'_{m,xy} \mu'_{m,y} + 4 (\mu'_{m,x})^2 \alpha'_{m,xx} \alpha'_{m,yy} \\ & + 4 (\mu'_{m,x})^2 \alpha'_{m,xx} \alpha'_{m,zz} + 8 (\mu'_{m,x})^2 (\alpha'_{m,xy})^2 \\ & - 12 \mu'_{m,x} \alpha'_{m,xy} \mu'_{m,y} \alpha'_{m,yy} - 4 \mu'_{m,x} \alpha'_{m,xy} \mu'_{m,y} \alpha'_{m,zz} \\ & + 8 (\mu'_{m,x})^2 (\alpha'_{m,xz})^2 - 8 \mu'_{m,x} \alpha'_{m,xz} \mu'_{m,y} \alpha'_{m,yz} + 9 (\mu'_{m,x})^2 (\alpha'_{m,yy})^2 \\ & + 6 (\mu'_{m,x})^2 \alpha'_{m,yy} \alpha'_{m,zz} + 12 (\mu'_{m,x})^2 (\alpha'_{m,yz})^2 + 9 (\mu'_{m,x})^2 (\alpha'_{m,zz})^2 \\ & + (\mu'_{m,z})^2 (9 (\alpha'_{m,xx})^2 + 6 \alpha'_{m,xx} \alpha'_{m,yy} + 4 \alpha'_{m,zz} (\alpha'_{m,xx} + \alpha'_{m,yy}) \\ & + 12 (\alpha'_{m,xy})^2 + 8 (\alpha'_{m,xz})^2 + 9 (\alpha'_{m,yy})^2 + 8 (\alpha'_{m,yz})^2 + 3 (\alpha'_{m,zz})^2) \\ & + 9 (\alpha'_{m,xx})^2 (\mu'_{m,y})^2 + 4 \alpha'_{m,xx} (\mu'_{m,y})^2 \alpha'_{m,yy} + 6 \alpha'_{m,xx} (\mu'_{m,y})^2 \alpha'_{m,zz} \\ & + 8 (\alpha'_{m,xy})^2 (\mu'_{m,y})^2 + 12 (\alpha'_{m,xz})^2 (\mu'_{m,y})^2 + 3 (\mu'_{m,y})^2 (\alpha'_{m,yy})^2 \\ & \left. + 4 (\mu'_{m,y})^2 \alpha'_{m,yy} \alpha'_{m,zz} + 8 (\mu'_{m,y})^2 (\alpha'_{m,yz})^2 + 9 (\mu'_{m,y})^2 (\alpha'_{m,zz})^2 \right) \end{aligned} \quad (S2)$$

$$\langle |e_a \underline{\mu}'_m|^2 |e_a \underline{\alpha}'_m e_b|^2 \rangle = \langle |e_a \underline{\mu}'_m|^2 |e_b \underline{\alpha}'_m e_a|^2 \rangle =$$

$$\begin{aligned} & \frac{1}{105} \left(-\alpha'_{m,xx} (-2 (\mu'_{m,x} \alpha'_{m,xy} \mu'_{m,y} + \mu'_{m,z} (\mu'_{m,x} \alpha'_{m,xz} - 2 \mu'_{m,y} \alpha'_{m,yz})) \right. \\ & + \alpha'_{m,yy} (3 (\mu'_{m,x})^2 + 3 (\mu'_{m,y})^2 + (\mu'_{m,z})^2) + \alpha'_{m,zz} (3 (\mu'_{m,x})^2 + (\mu'_{m,y})^2 + 3 (\mu'_{m,z})^2)) \\ & + (\alpha'_{m,xx})^2 (3 (\mu'_{m,x})^2 + 2 ((\mu'_{m,y})^2 + (\mu'_{m,z})^2)) + 6 \alpha'_{m,xy} \mu'_{m,z} (\mu'_{m,x} \alpha'_{m,yz} + \alpha'_{m,xz} \mu'_{m,y}) \\ & + 2 \mu'_{m,x} \alpha'_{m,xy} \mu'_{m,y} (\alpha'_{m,yy} - 2 \alpha'_{m,zz}) + (\alpha'_{m,xy})^2 (8 ((\mu'_{m,x})^2 + (\mu'_{m,y})^2) + 5 (\mu'_{m,z})^2) \\ & + 8 (\mu'_{m,x})^2 (\alpha'_{m,xz})^2 + 2 \mu'_{m,z} (\mu'_{m,x} \alpha'_{m,xz} (\alpha'_{m,zz} - 2 \alpha'_{m,yy}) + \mu'_{m,y} \alpha'_{m,yz} (\alpha'_{m,yy} + \alpha'_{m,zz})) \\ & + 6 \mu'_{m,x} \alpha'_{m,xz} \mu'_{m,y} \alpha'_{m,yz} + 2 (\mu'_{m,x})^2 (\alpha'_{m,yy})^2 - (\mu'_{m,x})^2 \alpha'_{m,yy} \alpha'_{m,zz} \\ & + 5 (\mu'_{m,x})^2 (\alpha'_{m,yz})^2 + 2 (\mu'_{m,x})^2 (\alpha'_{m,zz})^2 + 5 (\alpha'_{m,xz})^2 (\mu'_{m,y})^2 \\ & + (\mu'_{m,z})^2 (8 (\alpha'_{m,xz})^2 + 2 (\alpha'_{m,yy})^2 - 3 \alpha'_{m,yy} \alpha'_{m,zz} + 8 (\alpha'_{m,yz})^2 + 3 (\alpha'_{m,zz})^2) \\ & \left. + 3 (\mu'_{m,y})^2 (\alpha'_{m,yy})^2 - 3 (\mu'_{m,y})^2 \alpha'_{m,yy} \alpha'_{m,zz} + 8 (\mu'_{m,y})^2 (\alpha'_{m,yz})^2 + 2 (\mu'_{m,y})^2 (\alpha'_{m,zz})^2 \right) \end{aligned} \quad (S3)$$

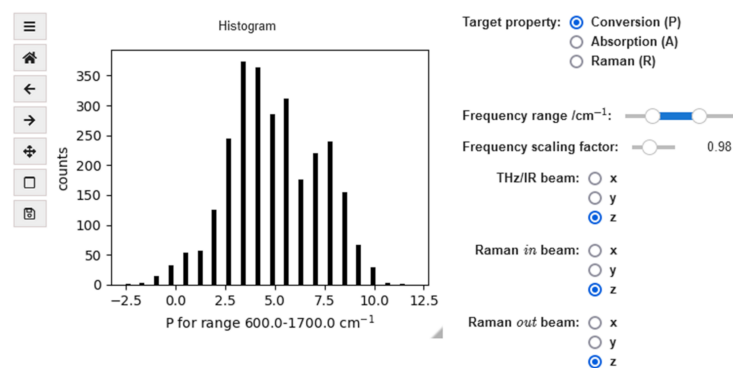
$$\langle |e_a \underline{\mu}'_m|^2 |e_b \underline{\alpha}'_m e_c|^2 \rangle =$$

$$\begin{aligned}
& \frac{1}{105} \left(-\alpha'_{m,xx} (4\mu'_{m,x} \alpha'_{m,xy} \mu'_{m,y} + 4\mu'_{m,z} (\mu'_{m,x} \alpha'_{m,xz} - 2\mu'_{m,y} \alpha'_{m,yz})) \right. \\
& + \alpha'_{m,yy} ((\mu'_{m,x})^2 + (\mu'_{m,y})^2 + 5(\mu'_{m,z})^2) \\
& + \alpha'_{m,zz} ((\mu'_{m,x})^2 + 5(\mu'_{m,y})^2 + (\mu'_{m,z})^2) \\
& + (\alpha'_{m,xx})^2 ((\mu'_{m,x})^2 + 3((\mu'_{m,y})^2 + (\mu'_{m,z})^2)) \\
& - 4\alpha'_{m,xy} (3\mu'_{m,z} (\mu'_{m,x} \alpha'_{m,yz} + \alpha'_{m,xz} \mu'_{m,y}) \\
& + \mu'_{m,x} \mu'_{m,y} (\alpha'_{m,yy} - 2\alpha'_{m,zz})) \\
& + (\alpha'_{m,xy})^2 (5((\mu'_{m,x})^2 + (\mu'_{m,y})^2) + 11(\mu'_{m,z})^2) \\
& + 5(\mu'_{m,x})^2 (\alpha'_{m,xz})^2 - 12\mu'_{m,x} \alpha'_{m,xz} \mu'_{m,y} \alpha'_{m,yz} \\
& + 4\mu'_{m,x} \alpha'_{m,xz} \mu'_{m,z} (2\alpha'_{m,yy} - \alpha'_{m,zz}) + 3(\mu'_{m,x})^2 (\alpha'_{m,yy})^2 \\
& - 5(\mu'_{m,x})^2 \alpha'_{m,yy} \alpha'_{m,zz} + 11(\mu'_{m,x})^2 (\alpha'_{m,yz})^2 \\
& + 3(\mu'_{m,x})^2 (\alpha'_{m,zz})^2 + 11(\alpha'_{m,xz})^2 (\mu'_{m,y})^2 + (\mu'_{m,z})^2 (5(\alpha'_{m,xz})^2 \\
& + 3(\alpha'_{m,yy})^2 - \alpha'_{m,yy} \alpha'_{m,zz} + 5(\alpha'_{m,yz})^2 + (\alpha'_{m,zz})^2) \\
& + (\mu'_{m,y})^2 (\alpha'_{m,yy})^2 - 4\mu'_{m,y} \alpha'_{m,yz} \mu'_{m,z} (\alpha'_{m,yy} + \alpha'_{m,zz}) \\
& \left. - (\mu'_{m,y})^2 \alpha'_{m,yy} \alpha'_{m,zz} + 5(\mu'_{m,y})^2 (\alpha'_{m,yz})^2 + 3(\mu'_{m,y})^2 (\alpha'_{m,zz})^2 \right) \tag{S4}
\end{aligned}$$

S2 Captions of the web application notebooks

Three interactive notebooks are available through our web application called Molecular Vibration Explorer. Their functionalities are described in the main text (Section IV - Functionalities) and screenshots are shown in Figs S1, S2, and S3.

Distribution of target property values in the database



Top molecules Page: 1

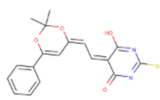
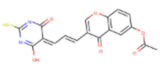
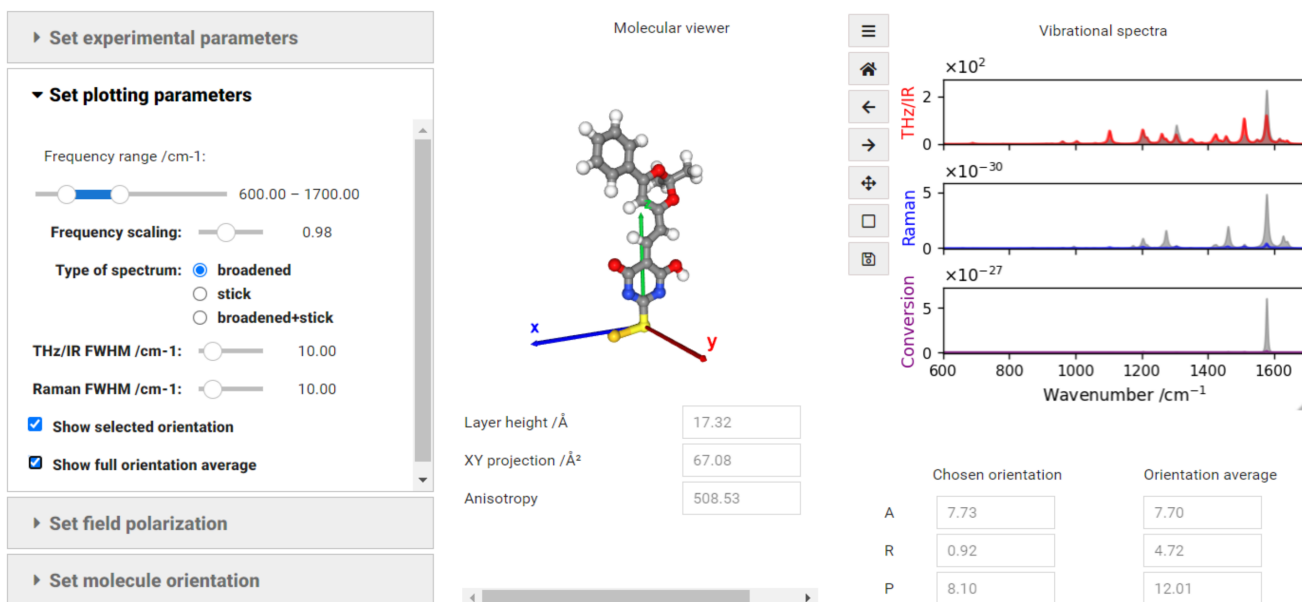
Index	Code	Molecule	P	A	R	Link to molecule page	Link to normal mode page
1	000-752-142		12.01	7.79	4.73	Go to molecule page	Check normal modes
2	000-754-182		11.26	7.44	4.68	Go to molecule page	Check normal modes

Figure S1: Notebook enabling the exploration of the Gold and Thiol databases. Accessed June 9, 2022.

Spectroscopic properties



SMILES code: SC1=NC(=O)/C=C/C=C/2\C=C(C(OC(O2)(C)C)c2ccccc2)/C(=N1)O
 Note on spectra units: THz/IR [km·mol⁻¹], Raman [cm²·sr⁻¹], Conversion [km·mol⁻¹·cm²·sr⁻¹]

Similar molecules

▶ Similar molecules in the database
▶ Similar self-assembly materials

Figure S2: Notebook displaying the molecular properties and vibrational spectra of a single molecule. Accessed June 9, 2022.

Normal modes

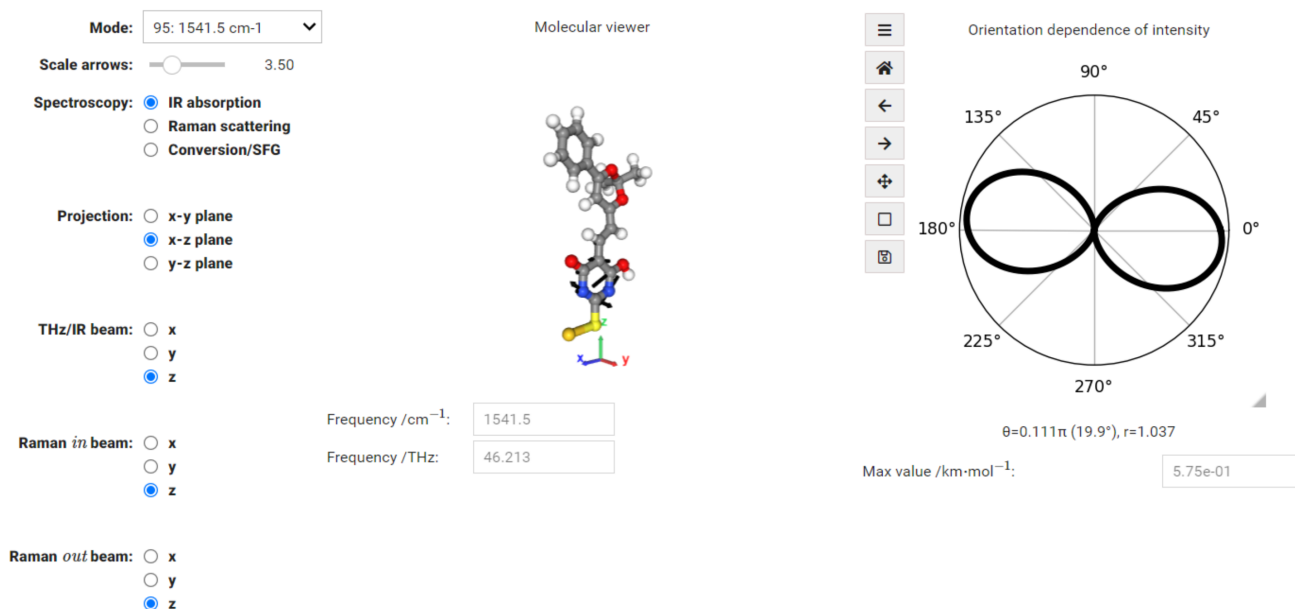


Figure S3: Notebook for the exploration of the different normal modes of the selected molecule. Accessed June 9, 2022.

S3 Comparing the *Thiol* and the *Gold* database

Thiolated and gold-linked versions of a molecule can be compared by a simple SMILES search and navigating the corresponding pages. The example in Fig. S4 shows orientation-averaged spectra and two vibrational modes, one enhanced considerably by linkage to a gold atom, and one that is active in both versions of the molecule.

SMILES search: N#CCc1c(S)cc(cc1Br)Br

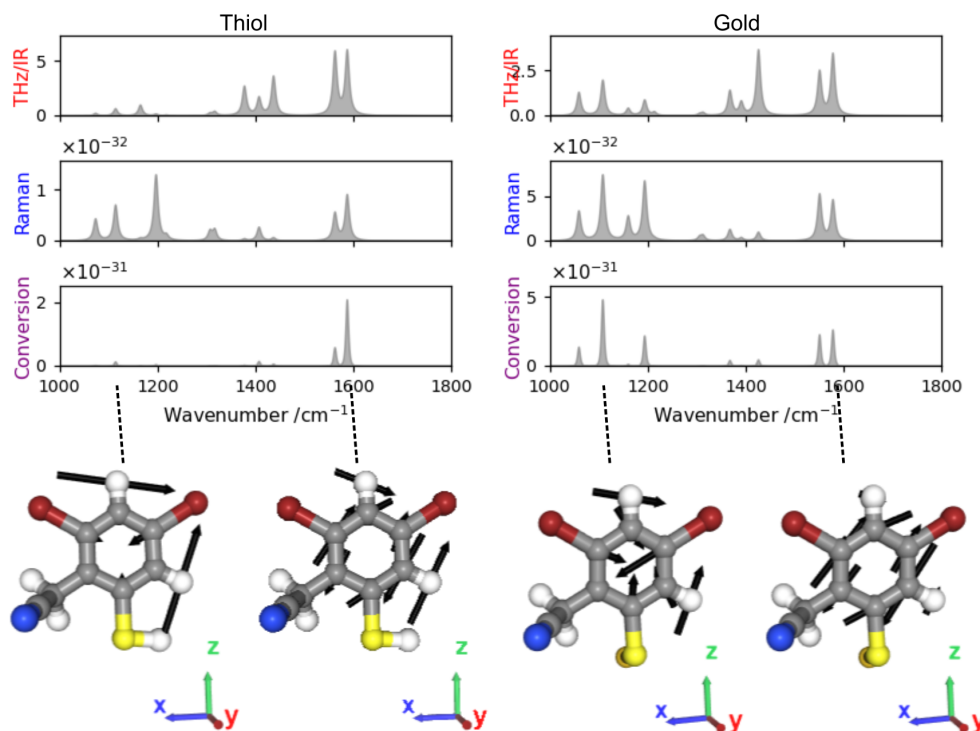


Figure S4: Using MVE to compare thiolated and gold-linked versions of a molecule.

Fig. S5 shows a comparison to experimental SERS and solution phase Raman data from Ref. 1.

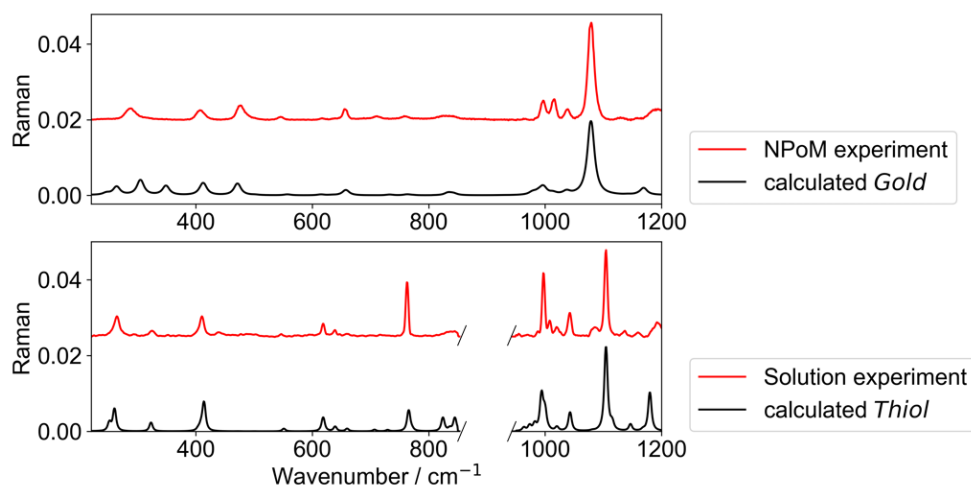


Figure S5: Raman spectrum of BPhT from SERS nanoparticle-on-mirror (NPoM) and solution (solvent: THF) measurements compared to calculations from the *Gold* and *Thiol* databases, respectively. Experimental data was taken from Ref. 1. Intensities are normalized for the integral of the spectra.

S4 List of molecules appearing in the manuscript

Table S1: Full names and SMILES of molecules appearing in the manuscript.

Name manuscript	IUPAC name	SMILES
NH2-BPhT	4-(4-aminophenyl)benzenethiol	<chem>C1=CC(=CC=C1C2=CC=C(C=C2)S)N</chem>
BPhT	[1,1'-biphenyl]-4-thiol	<chem>Sc1ccc(cc1)-c1ccccc1</chem>
Br-BPhT	4-(4-bromophenyl)benzenethiol	<chem>C1=CC(=CC=C1C2=CC=C(C=C2)Br)S</chem>
F-BPhT	4'-fluoro-[1,1'-biphenyl]-4-thiol	<chem>Fc1ccc(cc1)-c1ccc(S)cc1</chem>
Compound I	(4E)-4-[[4-(dimethylamino)phenyl]methylidene]-1-phenyl-2-sulfanyl-4,5-dihydro-1H-imidazol-5-one	<chem>CN(C)c1ccc(\C=C2\N=C(S)N(C2=O)c2ccccc2)cc1 t:9 </chem>
Compound II	(5E)-5-{2-[(4E)-2,2-dimethyl-6-phenyl-2,4-dihydro-1,3-dioxin-4-ylidene]ethylidene}-6-hydroxy-2-sulfanyl-4,5-dihydropyrimidin-4-one	<chem>CC1(C)O\C(=C\C=C2/C(O)=NC(S)=NC2=O)C=C(O1)c1ccccc1 c:9,12,17 </chem>
Compound III	N'-[[[(5Z)-6-hydroxy-4-oxo-2-sulfanyl-4,5-dihydropyrimidin-5-ylidene]methyl]-N-(4,6,7-trimethylquinazolin-2-yl)]guanidine	<chem>Cc1cc2nc(NC(=N)N\C=C3\C(O)=NC(S)=NC3=O)nc(C)c2cc1C c:13,16 </chem>
Compound IV	(5E)-6-hydroxy-5-[[[5-(phenylsulfanyl)furan-2-yl]methylidene]-2-sulfanyl-4,5-dihydropyrimidin-4-one	<chem>OC1=NC(S)=NC(=O)\C1=C\c1ccc(Sc2ccccc2)o1 c:4,t:1 </chem>
CBP	9-[4'-(9H-carbazol-9-yl)-[1,1'-biphenyl]-4-yl]-9H-carbazole	<chem>c1ccc2c(c1)n(-c1ccc(cc1)-c1ccc(cc1)-n1c3ccccc3c3ccccc13)c1ccccc21</chem>
VX nerve agent	N-[2-[ethoxy(methyl)phosphoryl]sulfanylethyl]-N-propan-2-ylpropan-2-amine	<chem>CCOP(=O)(C)SCCN(C(C)C)C(C)C</chem>
Compound V	2-{5-[[[(5Z)-1-methyl-4,6-dioxo-2-sulfanyl-1,4,5,6-tetrahydropyrimidin-5-ylidene]methyl]furan-2-yl]benzoic acid	<chem>CN1C(S)=NC(=O)\C(=C\c2ccc(o2)-c2ccccc2C(O)=O)C1=O c:3 </chem>
Compound VI	(5Z)-6-hydroxy-5-[[[5-(4-methyl-3-nitrophenyl)furan-2-yl]methylidene]-2-sulfanyl-4,5-dihydropyrimidin-4-one	<chem>Cc1ccc(cc1[N+](=[O-])=O)-c1ccc(\C=C2\C(O)=NC(S)=NC2=O)o1 c:18,21 </chem>
Compound VII	(5E)-1-(3,4-dimethylphenyl)-5-[[[4-fluorophenyl]amino]methylidene]-2-sulfanyl-1,4,5,6-tetrahydropyrimidine-4,6-dione	<chem>Cc1ccc(cc1C)N1C(S)=NC(=O)\C(=C/Nc2ccc(F)cc2)C1=O c:11 </chem>

References

- (1) Koczor-Benda, Z.; Boehmke, A. L.; Xomalis, A.; Arul, R.; Readman, C.; Baumberg, J. J.; Rosta, E. Molecular Screening for Terahertz Detection with Machine-Learning-Based Methods. *Phys. Rev. X* **2021**, *11*, 041035.